

B.Tech. Degree VI Semester Special Supplementary Examination January 2019

ME 15-1601 DYNAMICS OF MACHINERY (2015 Scheme)

Time : 3 Hours

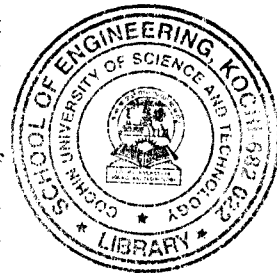
Maximum Marks : 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

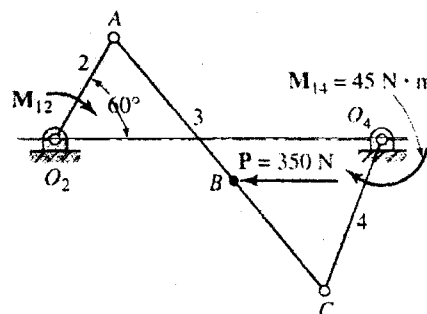
- I. (a) A horizontal engine has crank length 60 mm, connecting rod 240 mm and piston diameter 100 mm. When the crank has turned 30° from the inner dead centre, what will be the turning moment on the crank shaft due to a gas pressure of 580 kN/m^2 .
- (b) A rod of mass 10 kg and length 1m, has a radius of gyration of 300 mm about its centre of gravity, G. Point G is at a distance of 400 mm from one end. If this rod is to be replaced by a dynamically equivalent system consisting of a 6 kg mass and a 4 kg mass, find the distances to these masses from G.
- (c) The rim of a flywheel has a rectangular cross section, width 200 mm and thickness 40 mm. It has a mean diameter of 1.4 m. If the tensile stress, the rim material can withstand is 18 N/mm^2 , what is the maximum speed the flywheel can safely run? The rim material has a density of 7000 kg/m^3 .
- (d) Derive the expression for the ratio of tight side to slack side tensions for a band and block brake having n blocks.
- (e) A wheel of mass 6 kg and radius of gyration 250 mm, spinning at 300 rpm, is stopped using a block brake in 30 seconds. What will be the heat generated in the braking process?
- (f) There are three masses A, B and C revolving in a plane perpendicular to a shaft. They have masses 4 kg, 5 kg and 5 kg respectively and are attached to the shaft by arms which have lengths 300 mm, 400 mm and 300 mm respectively. Find the angle between the arms for complete balance.
- (g) Distinguish between static balancing and dynamic balancing.
- (h) What is the maximum torque that can be transmitted by the smaller pulley of 300 mm radius, if the angle of wrap of the belt around it is 160° , the initial tension in the belt is 1000 N, and the coefficient of friction between the belt and the pulley is 0.25.
- (i) In a chain drive, for constant angular velocity for the driving sprocket, will the angular velocity of driven sprocket remain constant? Why?
- (j) With a neat sketch explain how an inertia governor functions? What are its advantages over centrifugal governor?



PART B

(4 × 10 = 40)

- II. Determine the torque M_{12} to be applied to the link 2 of the four link mechanism shown in figure to maintain static equilibrium at the given position. Lengths $O_2A=200 \text{ mm}$, $AB=400 \text{ mm}$, $AC=700 \text{ mm}$, $O_2O_4=700 \text{ mm}$, $O_4C=350 \text{ mm}$.



OR

(P.T.O.)

- III. The connecting rod of a vertical high speed engine has a mass of 2.5 kg and is 30 cm long between centres. Its centre of gravity is 10 cm from the big end bearing. When suspended in a vertical plane and allowed to swing about the axis of the small end bearing, it makes 50 complete oscillations in 53 seconds. The stroke of the piston is 12.5 cm. The reciprocating parts have a mass of 1 kg. Determine the torque exerted on the crank shaft due to inertia of the moving parts when crank makes an angle of 140° with the top dead centre and the speed of rotation is 1600 rpm.

- IV. A shaft fitted with a flywheel rotates at 250 rpm and drives a machine. The torque of the machine varies in a cyclic manner over a period of 3 revolutions. The torque rises from 750 Nm to 3000 Nm uniformly during $\frac{1}{2}$ revolutions and remains constant for the following revolution. It then falls uniformly to 750 Nm during the next $\frac{1}{2}$ revolution and remains constant for one revolution, this being repeated thereafter. Determine the power required to drive the machine and percentage fluctuation of the speed, if the driving torque applied to shaft is constant and the mass of the flywheel is 500 kg and radius of gyration 60 cm.

OR

- V. A rail car having a total mass of 4000 kg is negotiating a curve of 300 m radius, where the track is banked at an angle of 10° . The car has two axles each of which together with its wheels and gearing has a total moment of inertia of 30 kg m^2 . The centre distance between the two wheels on an axle is 1.5 m and each wheel has 46.5 cm radius. Each axle is driven by a motor, the speed ratio between the two being 1:3. Each motor with its gear has a moment of inertia of 15 kg m^2 and runs in the direction opposite to that of the axle. The centre of gravity of the car is 105 cm above the rails when on level track. Determine the limiting speed for the car so that all the wheels maintain contact with the rail.

- VI. The intermediate cranks of a four cylinder symmetrical engine, which is in complete primary balance, are at 90° to each other and each has a reciprocating mass of 300 kg. The centre distance between the intermediate cranks is 700 mm and that between the extreme cranks is 1800 mm. Lengths of the cranks and connecting rods are 250 mm and 900 mm respectively. Calculate the masses fixed to the extreme cranks with their relative angular positions. Also find the secondary forces and couple if the engine runs at 600 rpm.

OR

- VII. Three cylinders of an air compressor have their lines of stroke 120° to one another and their connecting rods are coupled to a single crank. The stroke is 10 cm and the length of each connecting rod is 15 cm. Mass of the reciprocating parts per cylinder is 1.5 kg. Determine the primary and secondary unbalanced forces of the engine running at 3000 rpm.
- VIII. Power is transmitted by an open belt drive from a pulley 30 cm in diameter running at 200 rpm to a pulley 50 cm in diameter. Angle of lap on the small pulley is 160° . The belt is on the point of slipping when 2.6 kW is being transmitted. $\mu = 0.25$. It is desired to increase the power transmitted. State which of the following two methods suggested would be more effective: (i) Initial tension in the belt is increased by 10% (ii) Suitable dressing is given to the friction surface of the belt to increase the coefficient of friction by 10%.

OR

- IX. The mass of each ball of a Proell governor is 7.5 kg and the load on the sleeve is 80 kg. Each arm is 300 mm long. The upper arms are pivoted on the axis of rotation whereas the lower arms are pivoted to links of 40 mm from the axis of rotation. The extensions of the lower arms to which the balls are attached are 100 mm long and are parallel to the governor axis at the minimum radius. Determine the equilibrium speeds corresponding to extreme radii of 180 mm and 240 mm.